H Roll No. .....

TMA–201

B. TECH. (SECOND SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

ENGINEERING MATHEMATICS-II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Solve :

$$\left(\frac{y}{x}\sec y - \tan y\right)dx + (\sec y \log x - x)\,dy = 0$$

(CO1)

OR

(b) Solve : (CO1)

$$(D^4 + 2a^2D^2 + a^4)y = 8\cos ax$$

P. T. O.

Scanned with OKEN Scanner

2. (a) Solve : (CO1)

$$(D^4+1)y=e^x\cos x$$

OR

(b) Solve : (CO1)

$$\frac{d^2y}{dx^2}-2\frac{dy}{dx}+y=xe^x\sin x$$

3. (a) Solve : (CO1)

$$x^2\frac{d^2y}{dx^2}-x\frac{dy}{dx}+y=\log x$$

OR

(b) Solve : (CO1)

$$x^2\frac{d^2y}{dx^2}-3\frac{dy}{dx}+2y=\frac{e^x}{1+e^x}$$

by using variation of parameters method.

4. (a) Draw the graph of the following periodic function and find its Laplace Transform : (CO2)

$$f(t)=\begin{cases} t & \text{for } 0<t\leq a \\ 2a-t & \text{for } a<t<2a \end{cases}$$

OR

(b) If : (CO2)

$$f(t)=\frac{1-\cos t}{t^2}$$

find the Laplace transform of $f(t)$.

5. (a) Using Laplace transform, find the solution of the initial value problem : (CO2)

$$y''-4y'+4y=64\sin 2t$$

$$y(0)=0$$

$$y'(0)=1.$$

OR

(b) Using Convolution Theorem find : (CO2)

$$L^{-1}\left\{\frac{s}{(s^2+1)(s^2+4)}\right\}$$

Scanned with OKEN Scanner